Name: _____

# Observations

❑ One of the most important skills in Science is making observations. Observations are things we notice by using our senses.

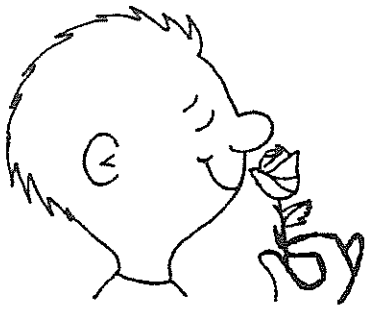
We use our eyes to _____

We use our ears to _____

We use our hands to _____

We use our noses to _____

We use our tongues to _____



❑ Imagine you are asked to observe the following. Which sense would you use in each case?

A dog's bark

A flower's scent

The colour of a car

The heat from a candle flame

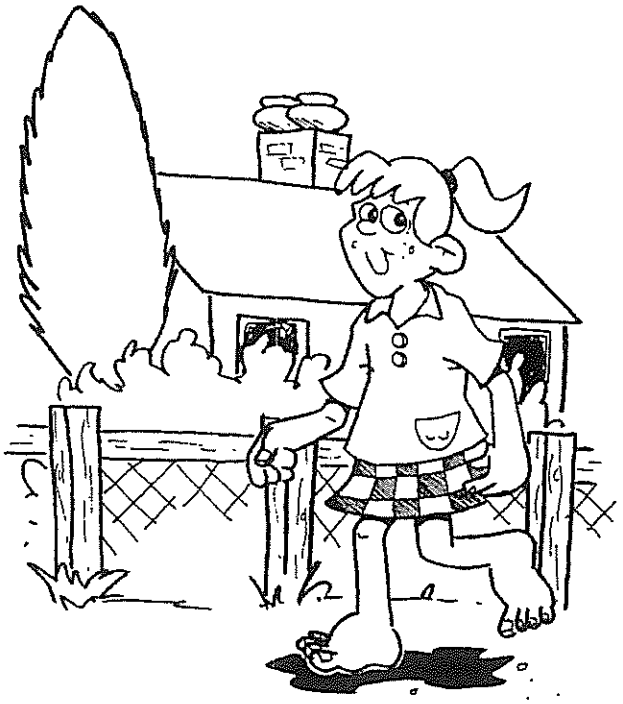
The flavour of a lollipop

❑ As you are reading this story, underline anything that is an observation.

As Meg walked outside, she heard the neighbours' cat, Dissie, meowing. She looked over the fence and saw Dissie crouched down near the bright yellow daffodils, playing with a ball.

Meg passed through the front gate, the scent of her mother's roses following her down the footpath. She wished she'd worn her shoes, as the ground was hot on her feet, warmed by the February sun. In the distance she heard the familiar sound of an ice-cream van.

The jingle, jangle noise of coins in her pocket brought a smile to Meg's face as she realised she could afford to buy her favourite food - toffee flavoured ice-cream.



Name: _____

# Testing Your Powers of Observation

❑ Sit with a partner. From your pencil cases, take a total of ten items and place them on your desk. Remove everything else from the desk. You will also need something to cover the items.

Firstly, one person studies the items for two minutes. The person is not allowed to write any notes.

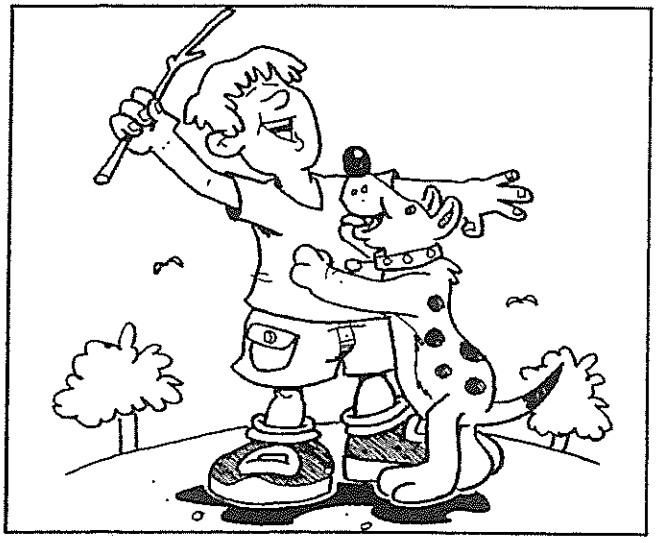
Next, the second person covers the items and the first person describes them in as much detail as possible, in the space below.

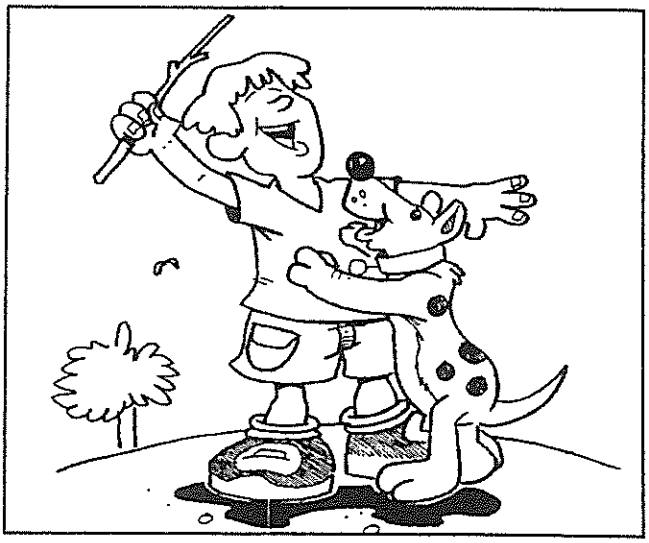
The ten items are...

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

Now swap your items with another group and the second person observes the items and tries to remember them.

❑ Look at the pictures below. Can you spot the ten differences? Circle them.





Name: _____

# How Well Do You Notice Things?

Often, when we describe objects, people or places, we just describe what we see.

Remember, that when you make observations, it is important to use all your senses if appropriate and if safe! For example, you might observe that bleach is a clear liquid that has a strong odour, but you wouldn't touch or taste it.

❑ **Look** around your classroom.
Make a list of all the blue objects.

_____

_____

_____

Make a list of all the objects made of wood.

_____

_____

_____

❑ Now, **listen** to your classroom.
Make a list of all the sounds you hear - try to describe how they sound.

_____

_____

_____

❑ Now, **smell** your classroom.
Make a list of all the odours you smell - try to describe how they smelt.

_____

_____

_____

❑ Close your eyes and carefully **feel** your way around some of your classroom.
Make a list of some of the things you felt - try to describe how they felt.

_____

_____

_____

Name: _____

# Totally Observant!

❑ For each of the following activities, list as many observations as possible. Try each activity by yourself first, then compare observations with other people in your class.

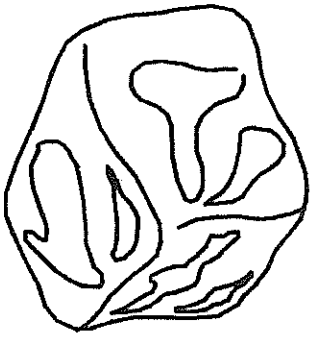
## Let There Be Light

Your teacher will give you a candle. Light it and note as many observations as you can.



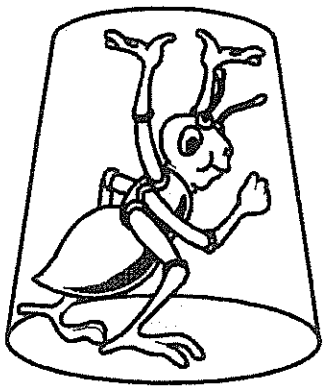
## Losing Shape

Put an iceblock in a tray. What do you notice? Remember to use all your senses.



## An Ant's Life

Put a glass over an ant so it cannot escape. Write down as many observations as you can. Take notice of any noises the ant makes, how it moves, its colour and so on.



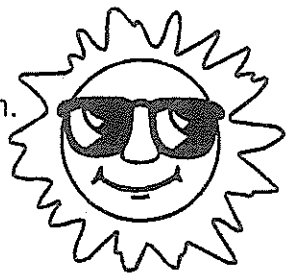
Name: _____

# Types of Observations

Observations can be made in two ways.

If the observation has no numbers it is called a *qualitative* observation.
For example: It is hot today.

If the observation has numbers it is called a *quantitative* observation.
For example: It is 34°C today.



❑ In the box under each observation, write the words 'quantitative' or 'qualitative' depending on which type it is.

The mug is heavy.

The mug weighs 550 grams.

The jug contains 850 millilitres of water.

The jug is full.

The train is late.

The train is eight minutes late.

The room is 10 m long.

The room is long.

Quantitative observations are also called *measurements*.
Measurements have a number value and a unit.

❑ In groups, discuss the units for the following measurements. There may be more than one unit for each. You can write your suggestions down here.

Time can be measured in: _____

_____

Mass (how heavy an object is) can be measured in: _____

_____

Volume (how much space an object fills) can be measured in: _____

_____

Temperature (how hot an object is) can be measured in: _____

_____

Length (how long an object/distance is) can be measured in: _____

_____

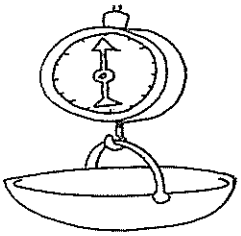
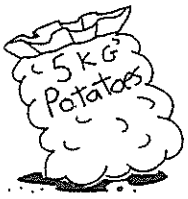
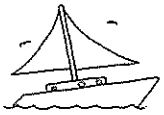
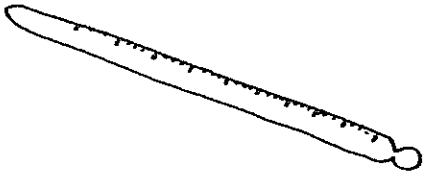
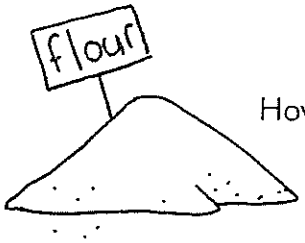
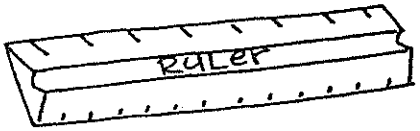
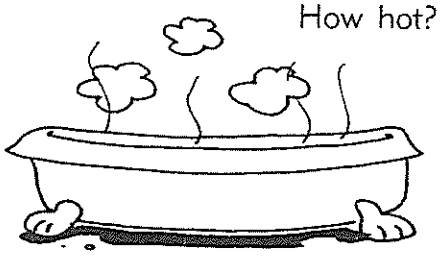
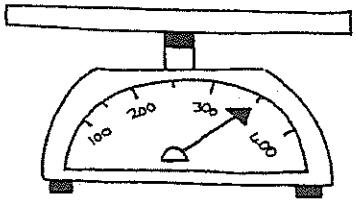
Name: _____

# Making Measurements

When we make measurements, we often use special instruments.

❑ Match the type of measurement with the instrument used by joining the pictures. Once you've finished, write the capital letter next to each picture in the corresponding space below. If you do it correctly, you should find the answer to this question.

**"What is the name of a large, greenish, Australian freshwater food fish?"**

| | | | | |
|---|---|---|---|---|
| 1 |  | How long? | M |  |
| 2 |   | How heavy? | R |   |
| 3 |  | How long? | Y |  |
| 4 |   | How heavy? | A |  |
| 5 |  | How much? | R |   |
| 6 |  | How hot? | U |   |

| 2 | 4 | 1 | 5 | 3 | 6 |
|---|---|---|---|---|---|
| | | | | | |

The large, greenish, Australian freshwater food fish is the _____ cod.

Name: _____

# Non-Fishy Scales

You may have used some of the instruments that measure things. You probably know how to use a ruler and a measuring jug.

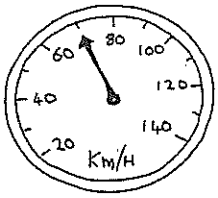
All measuring instruments have scales that divide the units so they are easier to read.

❑ Below are some measuring instruments. The arrow on each one indicates the value of the measurement.

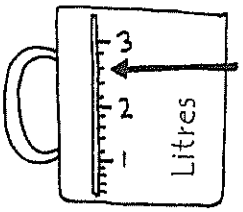
Once you have determined the correct value, write the correct units next to it ('g' for grams, 'cm' for centimetres and so on). Then, use a ruler to draw a straight line, joining the dots, and passing through the letter in the circles. Write the letters in the corresponding boxes. If you do it correctly, you should find the answer to this question.

**What is the name of a very light gas?**

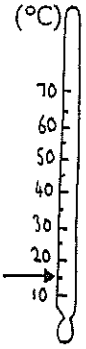
kilometres per hour (km/h)



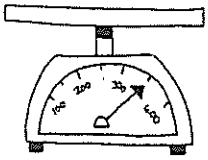

litres (L)



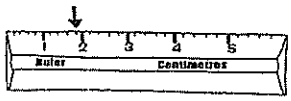

Celsius (°C)



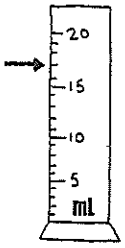
grams (g)



centimetres (cm)




millilitres (mL)




| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|

(L) (M) (E) (I) (H) (U)

| 17 | 70 | 350 | 2.6 | 15 | 1.8 |
|---|---|---|---|---|---|

| 5 | 4 | 1 | 6 | 3 | 2 |
|---|---|---|---|---|---|
| | | | | | |

The name of the very light gas is _____.

Name: _____

# Data - 1

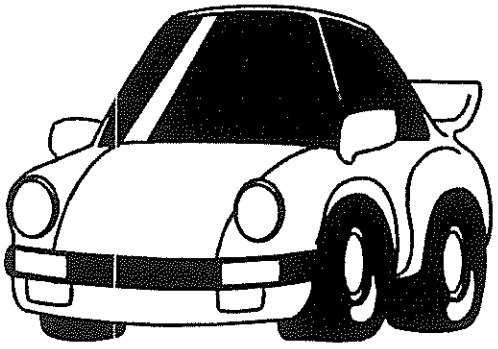
A group of observations or measurements is called *data*. Data is information that scientists can use to help explain things and to help answer questions.

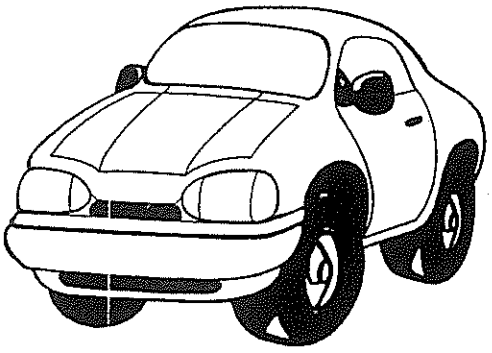
When scientists collect data, they can arrange it in a number of ways so it is easier to understand.

❑ Now, you will collect information (data) and arrange it in a table and a graph. Your teacher will take you outside to do this activity. You will need this worksheet and a pen.

1. Go to the school carpark.
2. Count the number of white cars, and write this number down in the space in the table below.
3. Do the same for the blue cars.
4. Continue counting the number of cars of each different colour until you have counted all the cars. If there are some cars in the carpark with colours that aren't listed in the table, you can add them in the spaces provided.

| Different colours of cars in our school's carpark on _____ | |
|---|---|
| **Colour of Car** | **Number of Cars** |
| White | |
| Blue | |
| Green | |
| Red | |
| Black | |
| Yellow | |
| Silver | |
| | |
| | |





a) How many red cars were there? _____

b) What colour is the most popular? _____

c) Were there more yellow cars or blue cars? _____

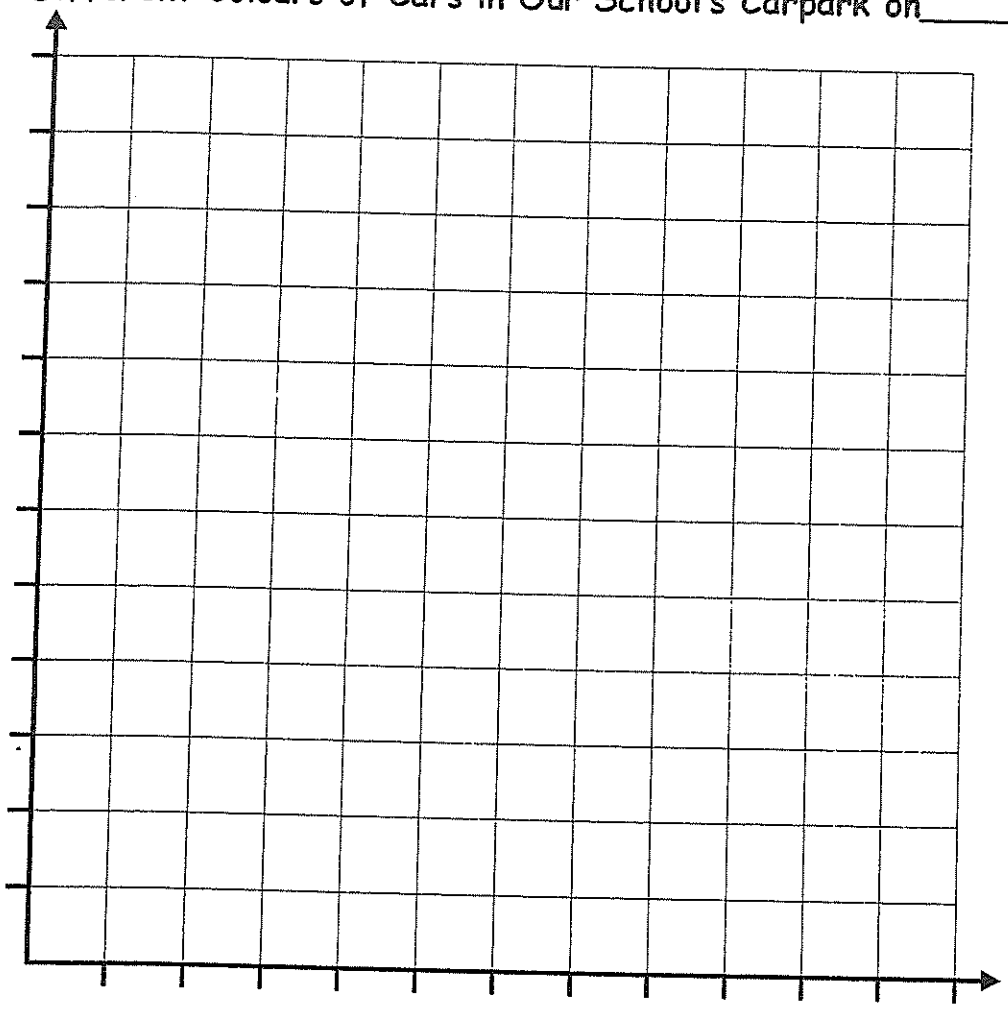
Name: _____

# Data - 2

❑ Now you are going to represent the data in another way. You are going to *graph* it. You will make a bar graph. Ask your teacher for help if you can't remember how to draw this type of graph.

Use a ruler to complete this graph.

**Different Colours of Cars in Our School's Carpark on_____**




**a)** How many green cars were there? _____

**b)** What colour is the least popular? _____

**c)** Were there more silver cars or more black cars? _____

You have used two different ways to show the data.

❑ Think about the questions that were asked.

Was it easier to use the table or the graph to answer the a) questions? _____

Was it easier to use the table or the graph to answer the b) questions? _____

Was it easier to use the table or the graph to answer the c) questions? _____

Scientists represent their data in the best way to answer their questions.



Educational Master

Name: _____

# How High is That Plant? - 1

Mrs Floral works in a nursery. Lots of her customers wanted to buy plants that will grow to be very tall. Mrs Floral knew of some tall plants but she didn't know which would be the tallest, so she decided to measure some of the plants in her nursery.

❑ Here are her results.

**Mrs Floral found that the daffodils grew to 40 cm, the November lilies reached 115 cm, the sunflowers were 180 cm and the gerberas grew to 70 cm.**

Name the four types of plants Mrs Floral has measured.

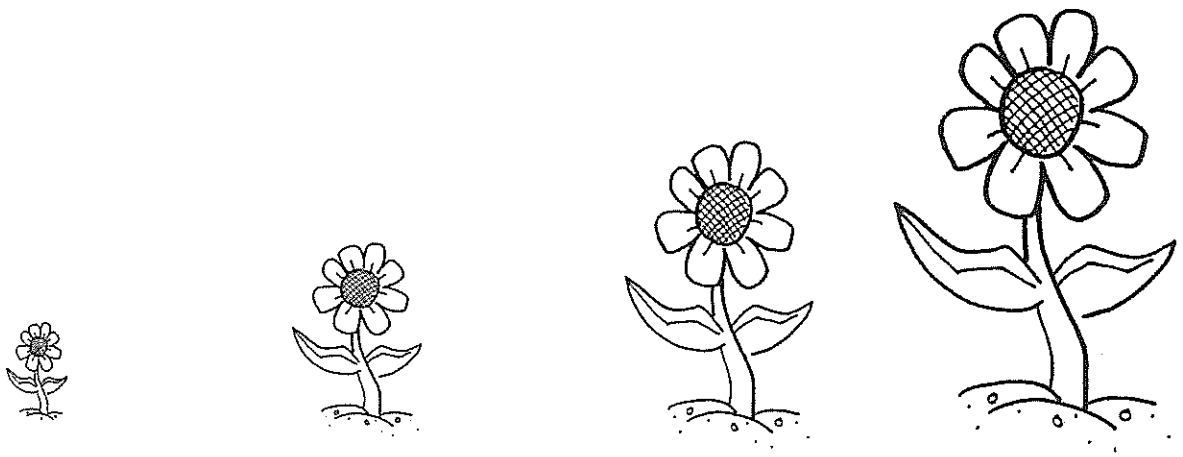
_____ _____

_____ _____

❑ 'Type of Plant' is one heading for your table. What do you think the other heading is? _____

Check this heading with your teacher then complete the table using the data Mrs Floral collected.

| | |
|---|---|
| | |
| | |
| | |
| | |
| | |



Name: _____

# How High is That Plant?- 2

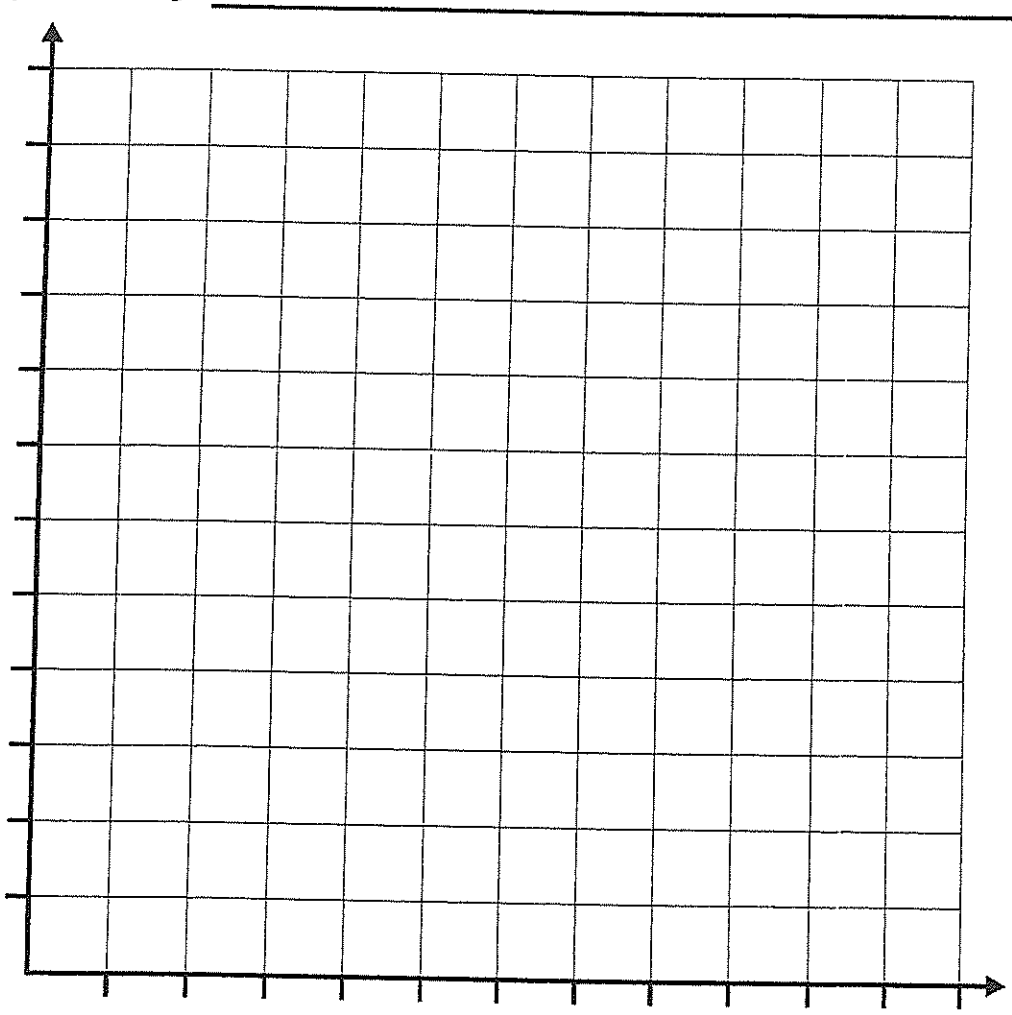
❑ Now you will draw a bar graph to show this information.

What will you put on the bottom axis of the bar graph? _____

What will you put on the side axis of the bar graph? _____

Don't forget to write these labels on your graph.

**Graph showing** _____



What plant grew to be the tallest? _____

What plant was the shortest? _____

Which plant was taller, the gerbera or the November lily? _____

Name: _____

# The Growing Metal - 1

Tristen's dad told him that if he heated a strip of copper metal with a strong flame, the strip would get longer. Tristen didn't believe his dad and decided to check for himself.

He heated a 50 cm strip of copper and checked its length every two minutes.

❑ These are his results:

After two minutes, the strip was 0.1 cm longer than before heating. It was 0.2 cm longer after four minutes, 0.3 cm longer after six minutes and 0.4 cm longer after eight minutes.

❑ Put this data in the table below.

| **Table showing the expansion (growth) of a copper strip** | | | | |
|---|---|---|---|---|
| Time (minutes) | 0 | | | |
| Extra length (cm) | 0 | | | |

Tristen thought he would be able to understand the results better if he put the data in a graph.

He decided to use a *line graph*. This type of graph is used when the data is *continuous*.

The data in the table is continuous because, even though Tristen didn't measure the strip of copper after one minute or five minutes, it must have still been expanding. It didn't just increase in length at two minutes, then again at four and six and so on. It was *continuously* increasing in length!

❑ Think back to the activity about the cars in the carpark. Was that data continuous? Explain your answer.

_____

_____

_____

_____

❑ Was the data about the height of Mrs Flora's plants continuous? Explain your answer.

_____

_____

_____

_____

Name: _____

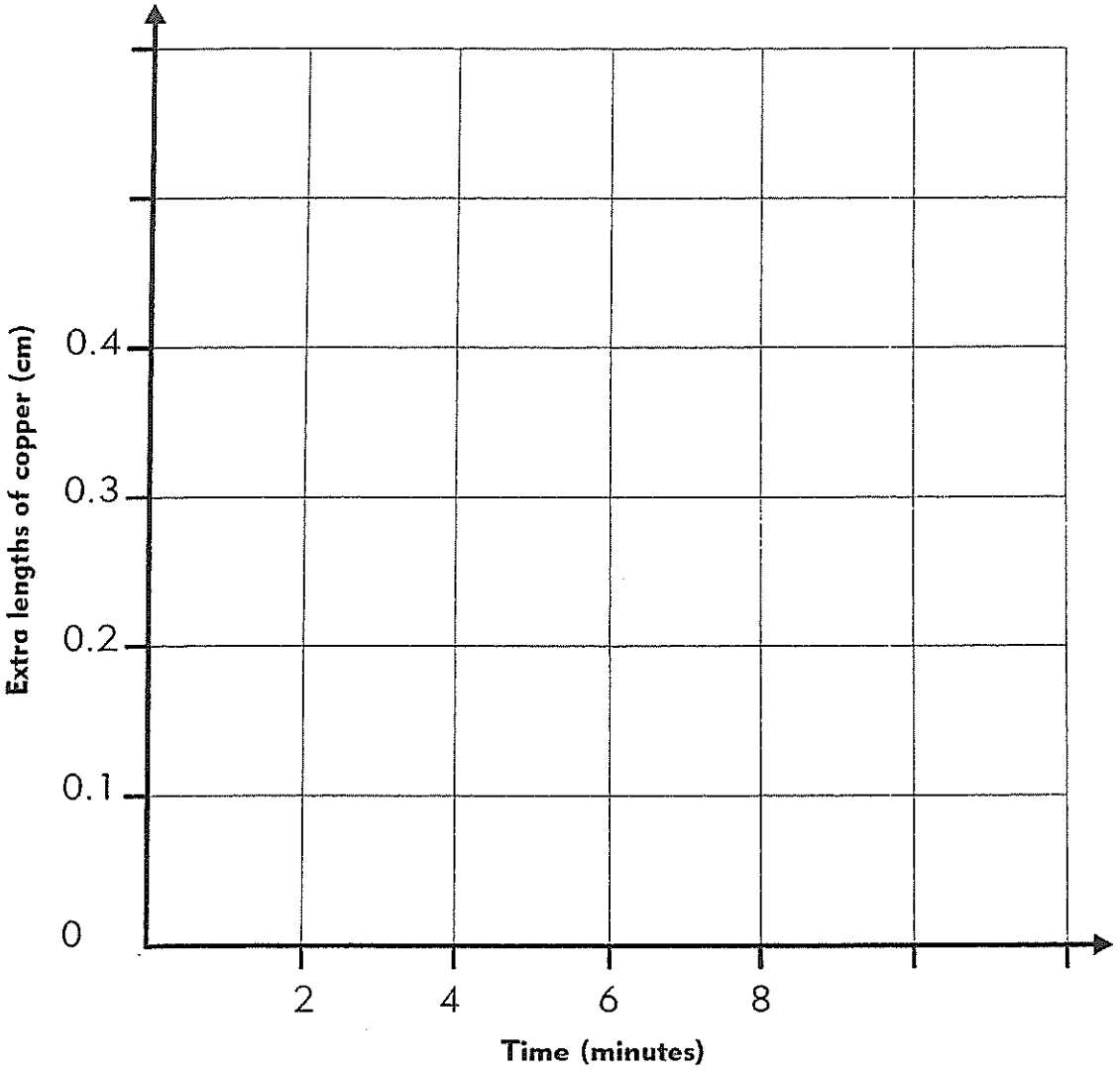
# The Growing Metal - 2

☐ Draw a line graph of the data Tristen collected.

Ask your teacher for help if you need it.

When you've marked all the points, join them up using a ruler.

**Graph Showing the Expansion of a Copper Strip**




☐ Use your graph to answer these questions.

**a)** What do you think the extra length of the copper strip would be after five minutes? _____

**b)** How long do you think it would take for the copper strip to have an extra length of 3.5 cm? _____

**c)** Use a ruler to determine the extra length of the copper strip after ten minutes, if it kept increasing in the same pattern. _____

Name: _____

# Tricky Line Graphs - 1

Two cars were driving to Mandurah. Every ten minutes, the distance covered by each car was recorded.

The data is shown in the table below.

| Table Showing Travel to Mandurah by Two Cars | | | | | | |
|---|---|---|---|---|---|---|
| Time (minutes) | 10 | 20 | 30 | 40 | 50 | 60 |
| Distance travelled by red car (km) | 15 | 30 | 45 | 60 | 75 | 90 |
| Distance travelled by blue car (km) | 20 | 40 | 60 | 80 | 100 | 120 |

This data is also continuous. Even though no measurements have been made for 12 minutes or 35 minutes, the cars were still driving at these times.

Once again, we will put this data into a line graph.

This data is more difficult to graph than that in the previous activity.

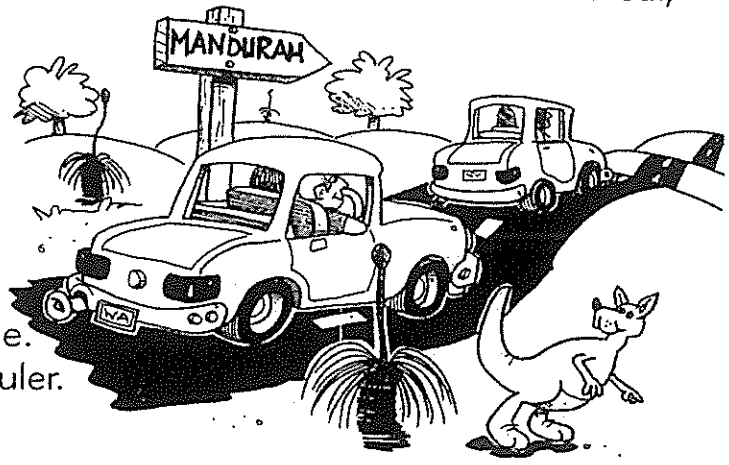
What makes it more difficult? _____

_____

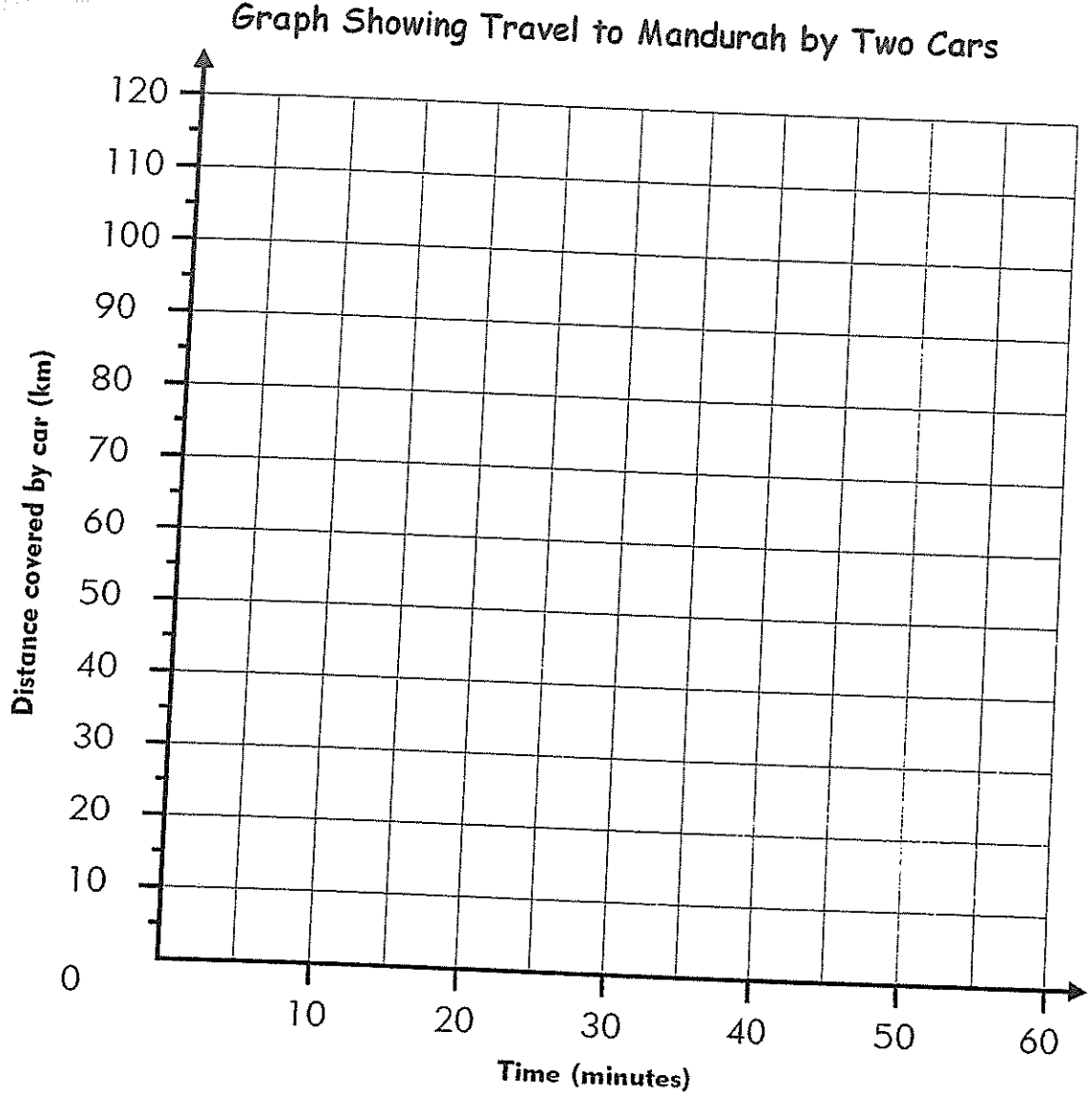
If you said it's because there are three lots of numbers instead of two, you'd be correct.

This means we will have two lines instead of one! One line will be for the red car, and the other will be for the blue car.

- ❑ Draw the line for the red car in red and the line for the blue car in blue. *(Don't forget to include a key/ legend and ask your teacher if you need help.)*
- ❑ Complete the graph on the next page. Remember to join the dots using a ruler.



# Tricky Line Graphs - 2




❑ Look at the table of data for the red car and the blue car.

Which car was driving fastest? _____

*(Hint: the car that has travelled the longest distance in a certain amount of time.)*

❑ Now look at your graph.

What do you notice about the car that has gone the fastest compared to the other car? _____

_____

*(Hint: look at the steepness of the line.)*

**By the way, the blue car was actually speeding (120 km/h) and was picked up and fined by the police!**

Name: _____

# Using Observations

If you notice a fire engine driving down your street, you could probably make several observations about it.

❑ List these observations here.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

If you wrote that you could HEAR the loud siren, or SEE the red fire engine, you have made observations.

But, if you said the fire engine is going to a fire, you have made an INFERENCE.

**An inference is a *possible explanation* of an observation.**

❑ Next to each of the following statements, write if you think it is an **observation** or an **inference**.

It is 32 degrees Celsius outside today. I think this is an _____

It is cold today because it is winter. I think this is an _____

The bird had blue feathers in its tail. I think this is an _____

The cake smelled of cinnamon. I think this is an _____

Jane is wearing a purple swimming costume because she is going swimming.

I think this is an _____

The car stopped because it ran out of petrol.

I think this is an _____



Name: _____

# More About Inferences

Did you notice something about your answers to the previous exercise?

The inferences all had the word 'because' in them. This is not always the case, but it is a good way to begin learning about observations and inferences.

☐ Below are some observations. You must give a possible explanation for each one. The first one has been done for you.

*Observation:* I heard a chainsaw outside.
*Possible Inference:* I heard the chainsaw because it was being used to cut down trees.

*Observation:* The bird was pecking the ground.
*Possible Inference:* The bird was pecking the ground because _____

*Observation:* The plant was withered.

*Possible Inference:* _____

*Observation:* The coffee was cold.

*Possible Inference:* _____

*Observation:* The dog was barking.

*Possible Inference:* _____

*Observation:* The car had the number 42 printed on its door.

*Possible Inference:* _____

*Observation:* The house smelt of bleach.

*Possible Inference:* _____

*Observation:* The cup of tea tasted sweet.

*Possible Inference:* _____

When you have finished, get in groups of three to discuss your inferences.

You may find that you all have different answers. Remember that an inference is only a *possible* explanation and there is often more than one possible explanation for any observation!

Name: _____

# But That's Not Fair!!

❑ Read the following scenarios.

**1.** Tom told Andrew that he thinks people who wear wetsuits float more easily than people who wear boardshorts and so they swim faster. Andrew disagreed, so they decided to test the idea. Tom put on his wetsuit, mask and snorkel, and fins and Andrew wore his purple boardshorts. They thought they'd test the idea by swimming one lap of a 25 metre pool. Tom couldn't dive in because he was wearing his mask so he jumped in. Andrew decided to dive.

❑ Tom won the race. Ruby, Andrew's sister, was watching and she said the race wasn't fair. List some of the reasons she may have given for saying this.

**2.** Mrs Rogers loves having flowers in vases on her kitchen table but she hates it when they die too fast. She decided to test some flowers to see which ones lasted the longest. She bought one bunch each of roses, carnations and daffodils. She put all the flowers in the same type of vase. Someone had told her that a bit of lemon juice would help the roses stay alive longer, so she added some to that vase. Someone else told her that the daffodils liked fresh air, so she moved that vase to the windowsill and kept the other two on the table. The carnations had lots of leaves and didn't fit in the vase properly, so she cut the leaves off.

❑ Do you think Mrs Rogers had set up a fair test? What would you have done to make it fair?

Name: _____

# Is That Fair?

Whenever you have a problem that you wish to solve, you can do a test. We call these tests experiments.

**If you do an experiment, it must be fair.**

For example, imagine you want to test who could eat two plates of food the fastest, you or your best friend. Would it be fair to get your friend to eat chewy steak while you ate watermelon? No, of course not! You would both have to eat the SAME food for the test to be fair.

❑ Now think, what other things (factors) would you have to keep the SAME to make it a fair test?

❑ Compare your ideas with those of some of your classmates and write any factors that they had and you did not.

THE ONLY FACTOR THAT YOU SHOULD CHANGE IS THE ONE YOU ARE TESTING.

In this case, THE PERSON EATING THE FOOD!

Name: _____

# Let's Settle This

Mr and Mrs Smart were having an argument. Mr Smart claimed that sugar dissolves more quickly than coffee in boiling water. Mrs Smart said the coffee dissolves more quickly. They decided to do an experiment to test their ideas.

The only factor they changed was the substance being added to the water (sugar or coffee).

Below is a list of the factors that they thought they would have to keep the same.

- The type of cup used,
- The temperature of the water.

If these were the ONLY factors they kept the same, was their experiment going to be fair? _____

❑ In groups of three, decide how to make it fair. Write your suggestions here.

_____

_____

_____

_____

❑ Draw a diagram below that shows exactly how they should set up the experiment.

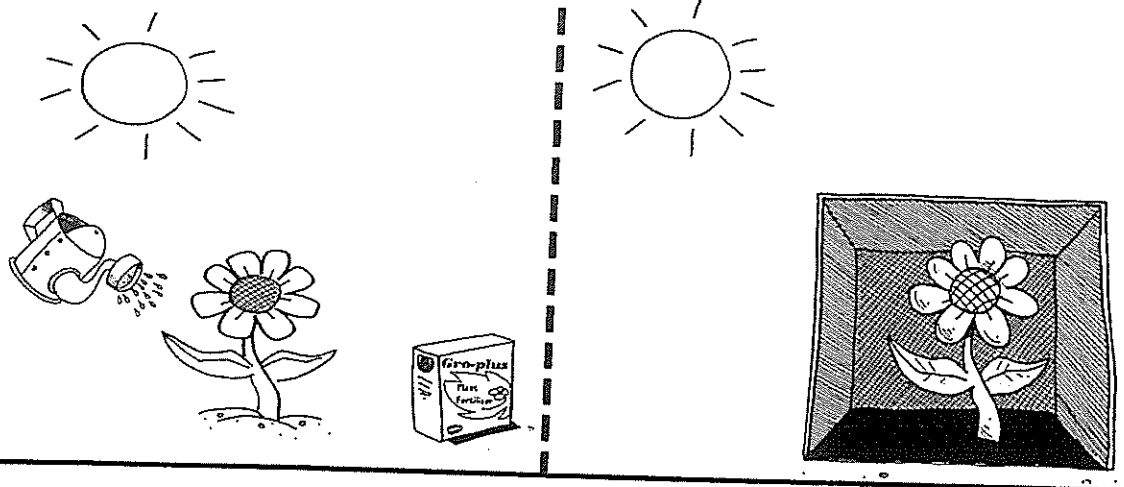
**Extension:** Try this experiment at home and report back to your class. Who was right, Mr or Mrs Smart?

Name: _____

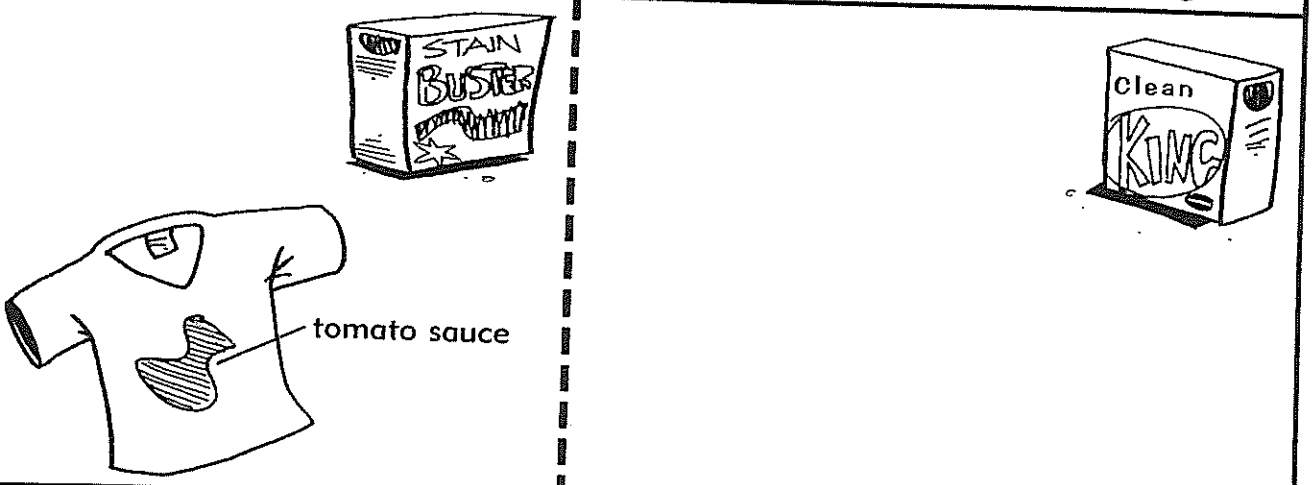
# More Fair Tests

❑ Here are some ideas to test.
For each one, finish the second picture to show how you could make it a fair test.

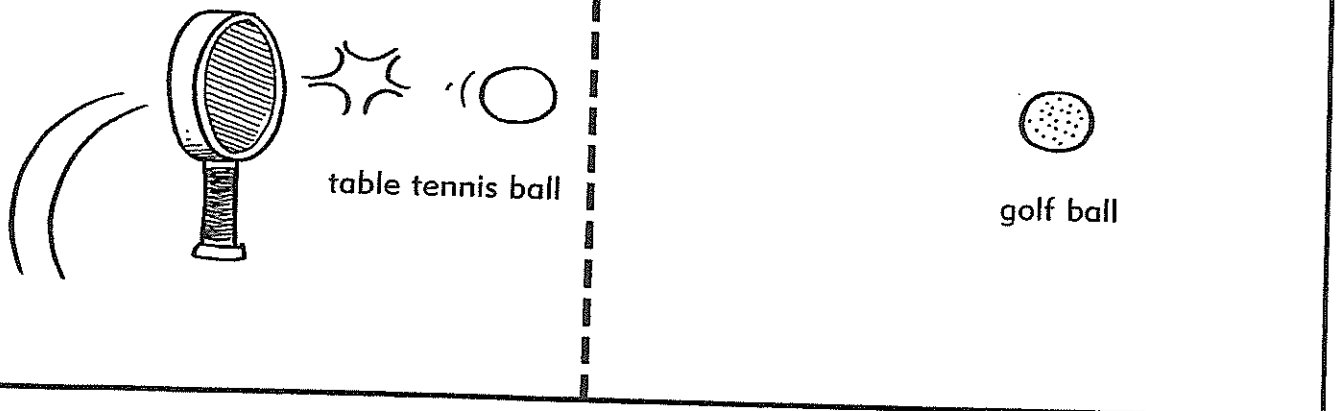
**Plants need sunlight to grow.**




**Stain Buster laundry powder removes stains better than Clean King.**




**A table tennis ball travels further than a golf ball when hit with a bat.**



Name: _____

# The Mystery of the Broken Torch

It's the middle of winter and a storm has cut off the electricity to your house. It's getting dark and you know you'll need light soon, so you find a torch and turn on the switch but nothing happens! The torch has a globe and batteries in it.

## What should you do?

❑ In groups, discuss how you would go about solving this problem. How will you prove why the torch isn't working?

Use these points to help you:

☆ What is the problem? _____

_____

☆ The torch didn't work because...
List all possible causes (inferences) of the problem.

_____

_____

_____

_____

_____

_____

☆ Choose one possible cause (inference) and use it to complete this statement...

*The torch didn't work because* _____

_____

☆ How are you going to test that this was the reason the torch wouldn't work? (Write out the steps you would take to test this inference.)

_____

_____

_____

_____

_____

_____

Name: _____

# Greenthumb?

You've planted some seedlings in the back yard. After a week, you notice that they look sick. After three weeks they are completely dead! You watered the plants everyday and used the same type of fertiliser on them. They were planted in the same soil and in the same place.

## What happened?

❑ In groups, discuss how you would go about solving this problem. How will you determine why the plants died?

Use these points to help you:

☆ What is the problem? _____

_____

☆ The plants died because...
List all possible causes (inferences) of the problem.

_____

_____

_____

_____

_____

_____

☆ Choose one possible cause (inference) and use it to complete this statement...

*The plants died because* _____

_____

☆ How are you going to test that this was the reason the plants died?
(Write out the steps you would take to test this inference.)

_____

_____

_____

_____

_____

_____

Name: _____

# Variables

## - A Fancy Word For Factors That Change!

Scientists have a special word for factors that **can** change in an experiment. They are called variables (from the word 'vary' meaning 'to change'). Remember that when you do an experiment, only one variable **should** change.

This factor is called the *test variable*. It is the variable being tested. In the case of the eating competition, the test variable is the *person eating*.

Another type of variable is called the *measured variable*. This is what is being measured.

In the case of the eating competition, the measured variable is the *time taken to eat the food*. The measured variable will change because the test variable changes.

The other variables, the ones that stay the same, are called *controlled variables*. In the case of the eating competition, the control variables are *the type of food eaten, the type of plate used, the location, the way the food is cut up*, and so on.

❑ Michael and Nicole had a competition to decide who could read the fastest.

Below are the variables they used. Next to each one, indicate if it is the test, measured or controlled variable/s.

The story to be read _____

The time taken to finish reading _____

The person reading _____

The room in which they read _____

The type of chair they sat on to read _____

❑ Mrs Floral did an experiment to test which type of plant grew the tallest.

Below are the variables she used. Next to each one, indicate if it is the test, measured or controlled variable/s.

The amount of water each plant received _____

The amount of sunlight each plant received _____

The type of plant _____

The type of soil used _____

The height of the plants _____

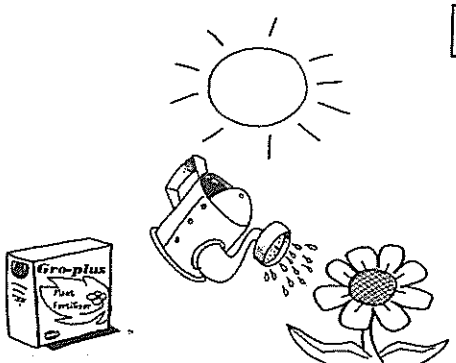
The type of fertiliser used _____

Name: _____

# Naming Variables

❑ Look at the pictures below. In a previous exercise you drew the second picture to make a fair test. Now, in the spaces provided, indicate whether each of the variable is the **test**, **measured** or **controlled** variables.

**Plants need sunlight to grow.**

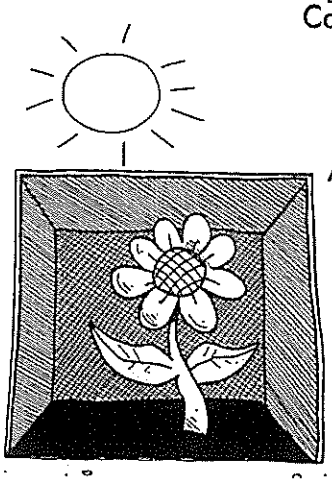



Sunlight _____

Condition of plant _____

Plant type _____

Amount and type of fertiliser _____

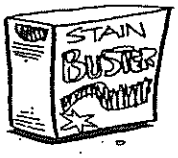




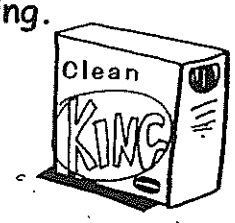

Amount of water _____

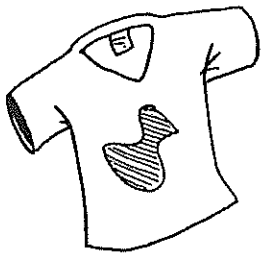
Soil type _____

**Stain Buster laundry powder removes stains better than Clean King.**

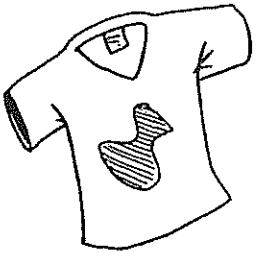



Type of powder _____



Method of washing _____



Disappearance of stain _____

Type of stain _____

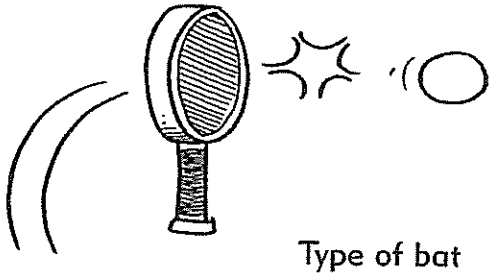
Type of T-shirt _____

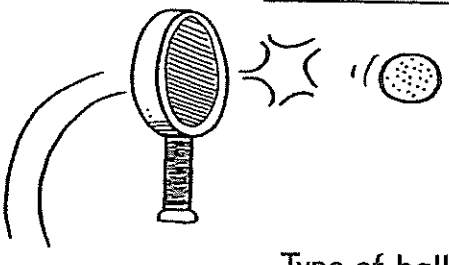
**A table tennis ball travels further than a golf ball when hit with a bat.**

Distance travelled by ball _____

Location _____

Strength of hit _____





Type of bat _____

Type of ball _____

Name: _____

# Testing Ideas

When we make an *educated guess* to explain an observation, it is called an **hypothesis**. An hypothesis is an inference that you can test.

For example, if you heard a baby crying (an observation), you might ask yourself "Why is the baby crying?" You might **infer** that she was hungry or tired.

If you then said "The baby will stop crying if I feed her" you are making an **hypothesis**. You can test the hypothesis by feeding the baby. If she stops crying, she must have been hungry! If she doesn't stop, she may have been tired.

What hypothesis would you write to test if the baby was tired?

_____

_____

_____

If you said "The baby will stop crying if I try to put her to sleep" then you've just written an hypothesis.

☐ See if you can write an hypothesis for the questions written below.

1. Does pure water boil at a lower or higher temperature than salted water?

_____

_____

_____

2. Why was the cat meowing at the back door?

_____

_____

_____

3. Which herb grows more quickly, parsley or thyme?

_____

_____

_____

4. Does the amount of water in a washing machine affect how clean the clothes get?

_____

_____

_____

Name: _____

# Hot or Cold? - 1

If you have a cup of tea, you know that various factors can affect how quickly a sugar cube will dissolve in it. It depends on how hot the water is, how often you stir, how much water is in the cup and so on. These are the possible variables.

You are now going to PLAN and then DO an **experiment**.

❑ In this experiment you are going to test if the temperature of the water affects how quickly a sugar cube will dissolve. This is called the **aim** of the experiment.

**Aim:** To test if the temperature _____

_____

_____

❑ Next, you have to make an educated guess at what will happen. This is called an **hypothesis** (hi-poth-e-sis). An hypothesis is like an inference that you test.

**Hypothesis:** The sugar cube will dissolve more quickly in hot / cold water.
(Circle your choice.)

❑ Next, you need to write a list of **materials** you will need to do the experiment.

**Materials:**
2 glasses of the same size
2 sugar cubes of the same size
2 teaspoons (or something else to stir with)
2 stopwatches
some hot water
some cold water

❑ Now, you are ready to plan the experiment. Remember that it has to be a fair test. The test variable is the temperature of the water - EVERYTHING else has to be kept the SAME. The measured variable is the time taken for the sugar cube to dissolve. This will change because the temperature of the water is changing.

❑ Write a sample plan (method) here. What are you going to do?

_____

_____

_____

_____

_____

_____

_____

_____

Name: _____

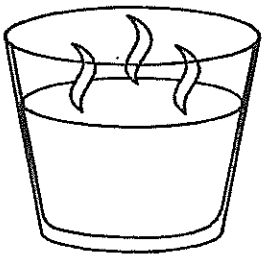
# Hot or Cold? - 2

How did you go? Check your method with the one written here. Your method may not be exactly the same but it should be similar.

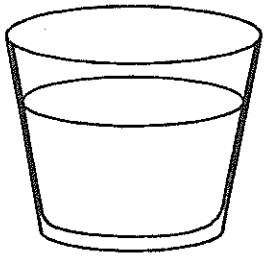
**Method:**

1. Half fill one glass with cold water and half fill the other glass with hot water.
2. Carefully place one sugar cube in each glass and start timing.
3. Stir each glass in the same way until the sugar has dissolved.
4. Record the time it takes for each sugar cube to dissolve.

☐ Finish the diagrams below showing what the two glasses will look like when they are set up.



HOT water



COLD water

☐ Do the experiment now, and write your findings (**results**) in this table.

**Results:**

| **The effect of temperature on dissolving sugar** | |
|---|---|
| Type of water in glass | Time for sugar cube to dissolve (min) |
| Cold water | |
| Hot water | |

☐ Finally, you need to write a brief summary of what happened. To do this, you need to re-read your aim and hypothesis. This summary is called a **conclusion**.

**Conclusion:**

It was found that the sugar cube dissolved more quickly in hot / cold water. (Circle your choice.)

Was your hypothesis right? _____

It doesn't matter if it was wrong, as long as you did a fair experiment!

Name: _____

# Big or Little? - 1

In this experiment, you will find out if the **size** of the pieces of sugar affects how quickly a sugar cube will dissolve.

**Aim:** To test if the _____

_____

**Hypothesis:** The sugar cube will dissolve more quickly if it is in big/small pieces. (Circle your choice.)

**Materials:**
2 glasses of the same size
2 teaspoons (or something else to stir with)
2 stopwatches
water
1 whole sugar cube (big piece)
1 crushed sugar cube (small pieces)

**Method:**

1. Half fill two glasses with water.

   *(Note:* will you use HOT or COLD water? Circle your choice.)

2. Carefully place one whole sugar cube in one glass and one crushed sugar cube in the other and start timing.

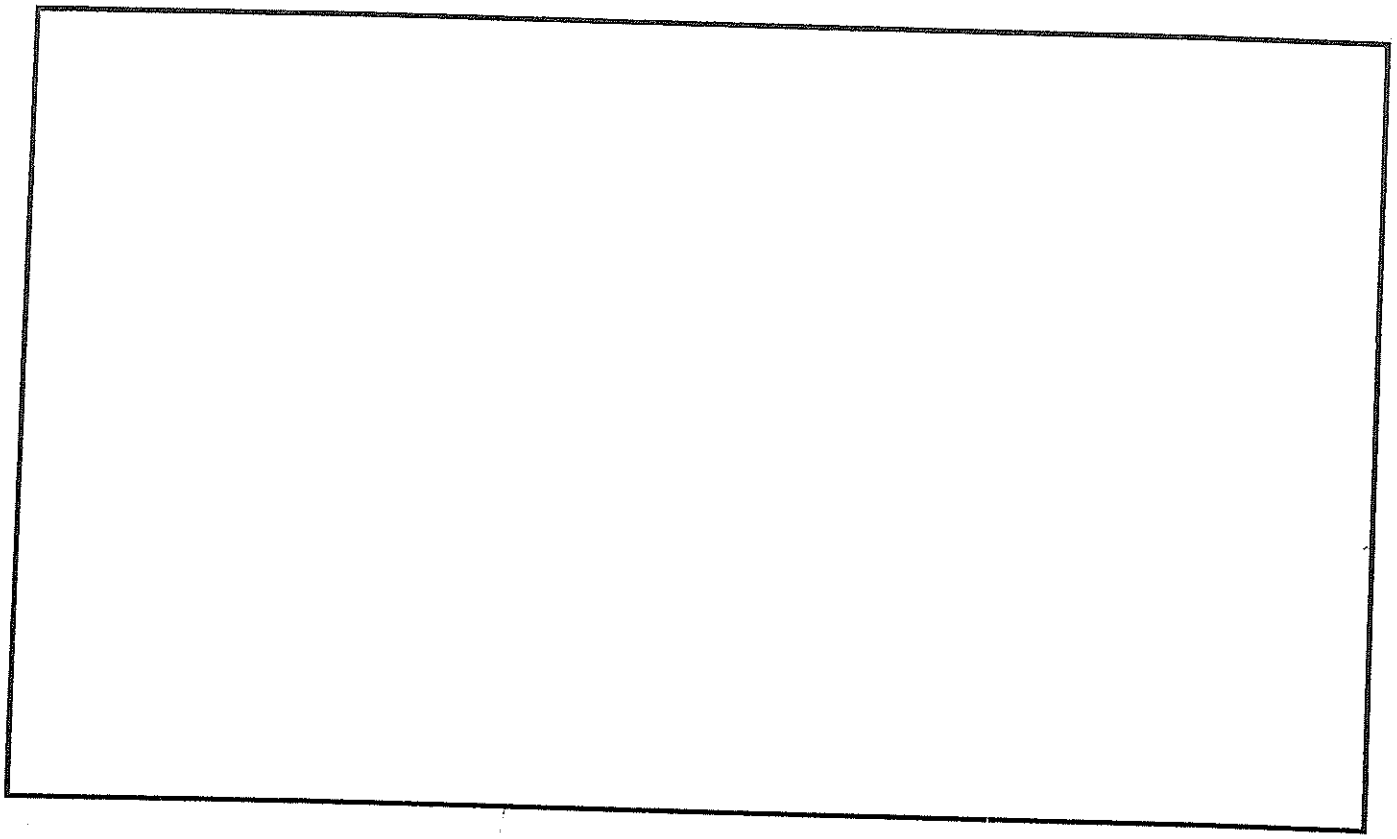
3. _____

4. _____

Name: _____

# Big or Little? - 2

## Diagram



## Results:

| The effect that the size of sugar has on how fast it dissolves | |
|---|---|
| Size of sugar | Time for sugar cube to dissolve (min) |
| Whole cube (big) | |
| Crushed cube (small) | |

## Conclusion:

It was found that the sugar cube dissolved more quickly when it was whole/crushed. (Circle your choice.)

What was the test variable in this experiment? _____

What was the measured variable? _____

What were the controlled variables? _____

_____

Name: _____

# To Stir Or Not To Stir?

❑ Next, you will find out if **stirring** affects how quickly a sugar cube dissolves.

Write the full report on file paper. Include all the sections you have used in the previous experiments.

Before you begin the experiment you must plan it. Answer these questions to help you do this.

What is the test variable? _____

What is the measured variable? _____

What are the controlled variables? _____

_____

_____

How much water will you use? _____

_____

What size of sugar particle will you use? (cube or crushed) _____

_____

How will you stir the sugar? _____

_____

_____

How will you decide when the sugar has been dissolved? _____

_____

_____

_____

_____

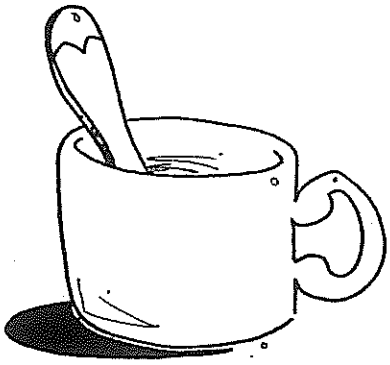
What data will you put in a table? _____

_____

_____

_____

_____



Don't forget to include a diagram. GOOD LUCK.

Name: _____

# What Makes Seeds Sprout? - 1

Have you ever wondered what makes seeds sprout? Maybe you already know some of the things (factors) they need.

❑ List the factors you think seeds need to sprout.

_____

_____

_____

_____

_____

_____

❑ You are going to do an experiment to see if seeds need soil to sprout.

**Aim:** To determine if _____

_____

_____

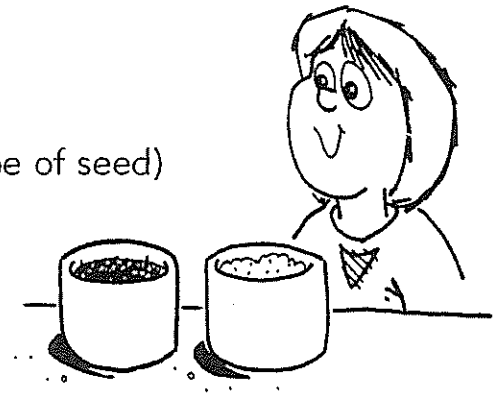
**Hypothesis:** Seeds will only sprout if they have soil.

OR

Seeds will sprout with or without soil.

❑ Underline the hypothesis you want.

**Materials:**
seeds (check with your teacher for the type of seed)
2 identical open containers
water
soil
cotton wool



**Method:**

1. Put soil in one of the containers and cotton wool in the other.
2. Sprinkle 20 seeds into each container
3. Water the seeds in each of the containers with the same amount of water. The soil should be moist.
4. Place both containers in the sunlight.
5. Add the same amount of water to both containers every day to keep them moist.
6. Record how many seeds have sprouted (in total) each day (you may not need 12 days).

Name: _____

# What Makes Seeds Sprout? - 2

**Diagram:**

**Results:**

| The effect that soil has on seeds sprouting | | |
|---|---|---|
| Day | Number of seeds that have sprouted | |
| | With soil | No soil |
| 0 | 0 | 0 |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |
| 9 | | |
| 10 | | |
| 11 | | |
| 12 | | |

Are there any other observations you should make? _____

_____

If seeds sprouted in 'no soil' do they look any different to those in soil? _____

_____

Describe your observations. _____

_____

_____

Name: _____

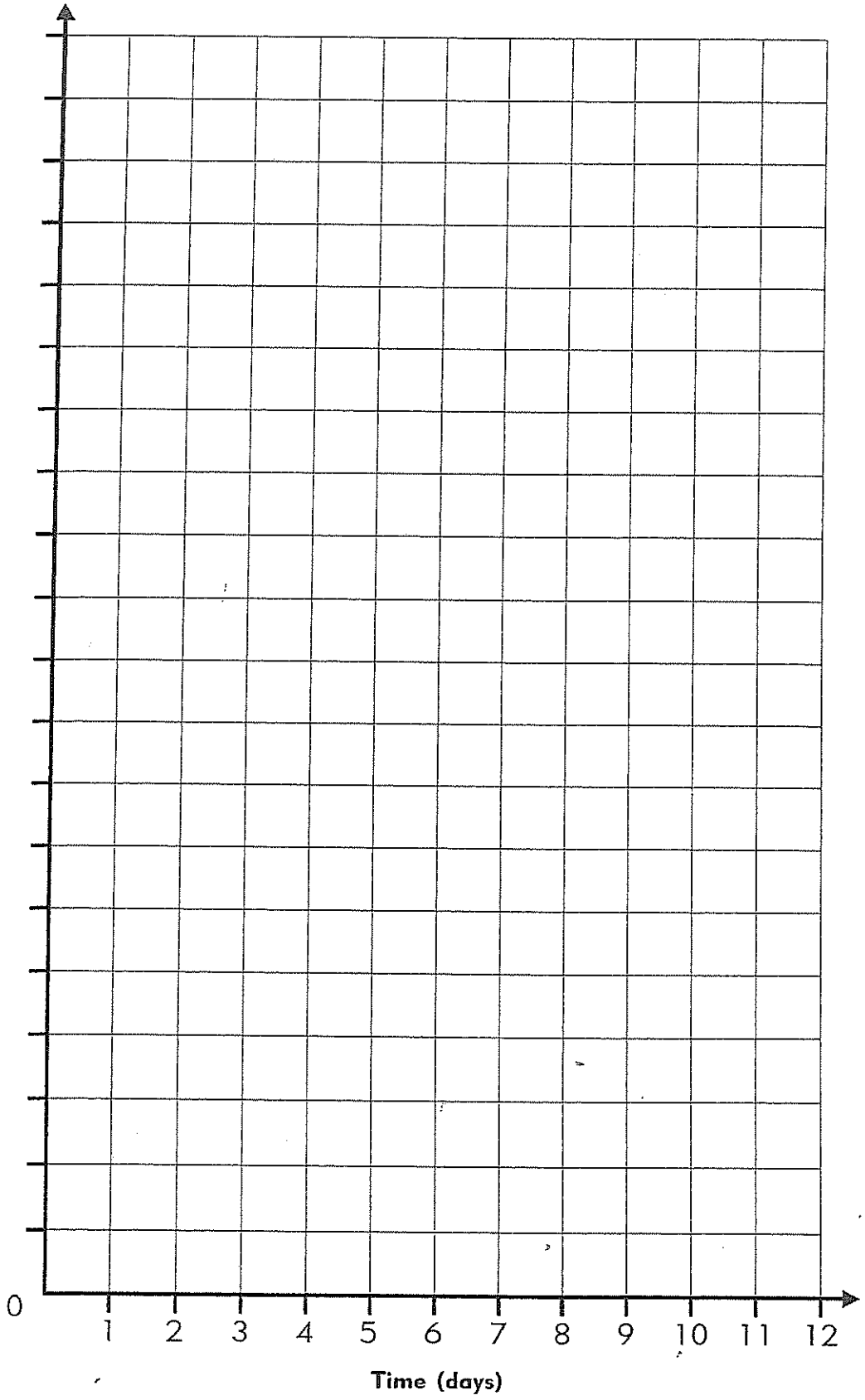
# What Makes Seeds Sprout? - 3

❑ On the grid below, you are to draw two line graphs in *different* colours.

Red = seeds with soil

Blue = seeds with no soil

**Graph showing the effect that soil has on seeds sprouting**



Name: _____

# What Makes Seeds Sprout? - 4

**Conclusion:** Seeds do / do not need soil to sprout.

What was the test variable in this experiment? _____

What was the "measured" variable in this experiment? _____

What were the "controlled" variables in this experiment? _____

Was this a fair experiment? _____

Discuss: Do you think your conclusion is true for all types of seeds? _____

What could you do to extend this experiment? _____

Name: _____

# What Do You Want to Test? - 1

☐ Choose one of the other factors that you thought seeds needed to sprout. Design an experiment to test if seeds do or do not need this factor.

**Aim:** To determine if _____

_____

**Hypothesis:** Seeds need / do not need _____

_____

What is your test variable? _____

What is your measured variable? _____

What are your control variables? _____

**Materials:** seeds (check with your teacher for the type of seed)
2 identical open containers
water
soil
*NOTE:* You may want to add other materials.

_____

_____

**Method:**

_____

_____

_____

_____

_____

_____

_____

**Diagram:**

Name: _____

# What Do You Want to Test? - 2

**Results:**

| The effect that _____ has on seeds sprouting | | |
|---|---|---|
| Day | Number of seeds that have sprouted | |
| | | |
| 0 | 0 | 0 |
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |
| 8 | | |
| 9 | | |
| 10 | | |
| 11 | | |
| 12 | | |

❑ Remember to describe any other observations you make.

**Conclusion:**

_____

_____

_____

_____

_____

Was your experiment fair? _____

_____

**It's all up to you now!**
On a separate sheet of paper, design an experiment to test one of the other factors that you think could affect the sprouting of a seed.

Name: _____

# Black or Silver - 1

Have you ever noticed that if you wear a black T-shirt in the sun, you get hotter than if you wear a white one?

- ☐ In this experiment, you are going to determine which colour will get hotter in sunlight, silver or black.
  You are to be a real scientist and design the experiment with your group. Write the completed experiment, neatly, on a separate sheet of paper.
- ☐ Remember to plan your experiment. Use this worksheet to help you.

What question are you trying to answer? This is your aim.

_____

_____

_____

What do you think will happen? This is your hypothesis.

_____

_____

_____

_____

What is your test variable? What is the one thing you are changing in this experiment?

_____

_____

_____

What is your measured variable? What will you measure?

_____

_____

_____

What are your controlled variables? Everything in your experiment must be the same, except your test variable. List all the things you must keep the same.

_____

_____

_____

_____

_____

Name: _____

# Black or Silver - 2

You can decide which silver and black objects you use, but you should check with your teacher to make sure they are available. You may wish to bring some items in from home. Also, ask your teacher if you can have a thermometer that measures up to 110°C. This is called a laboratory thermometer and may be useful for measuring how hot your objects get.

List the materials/equipment you think you might need here.

How will you do the experiment? This is your method. Remember a diagram.

How will you present your results? Remember that scientists like to use tables. Decide if you think a graph is needed.

Give this table the headings you will use. You may decide to include more information than will fit in this table. Just make sure that your table is easy to understand. Remember to include all observations.

| | |
|---|---|
| Black | |
| Silver | |

❑ Don't forget to write a conclusion to your experiment.